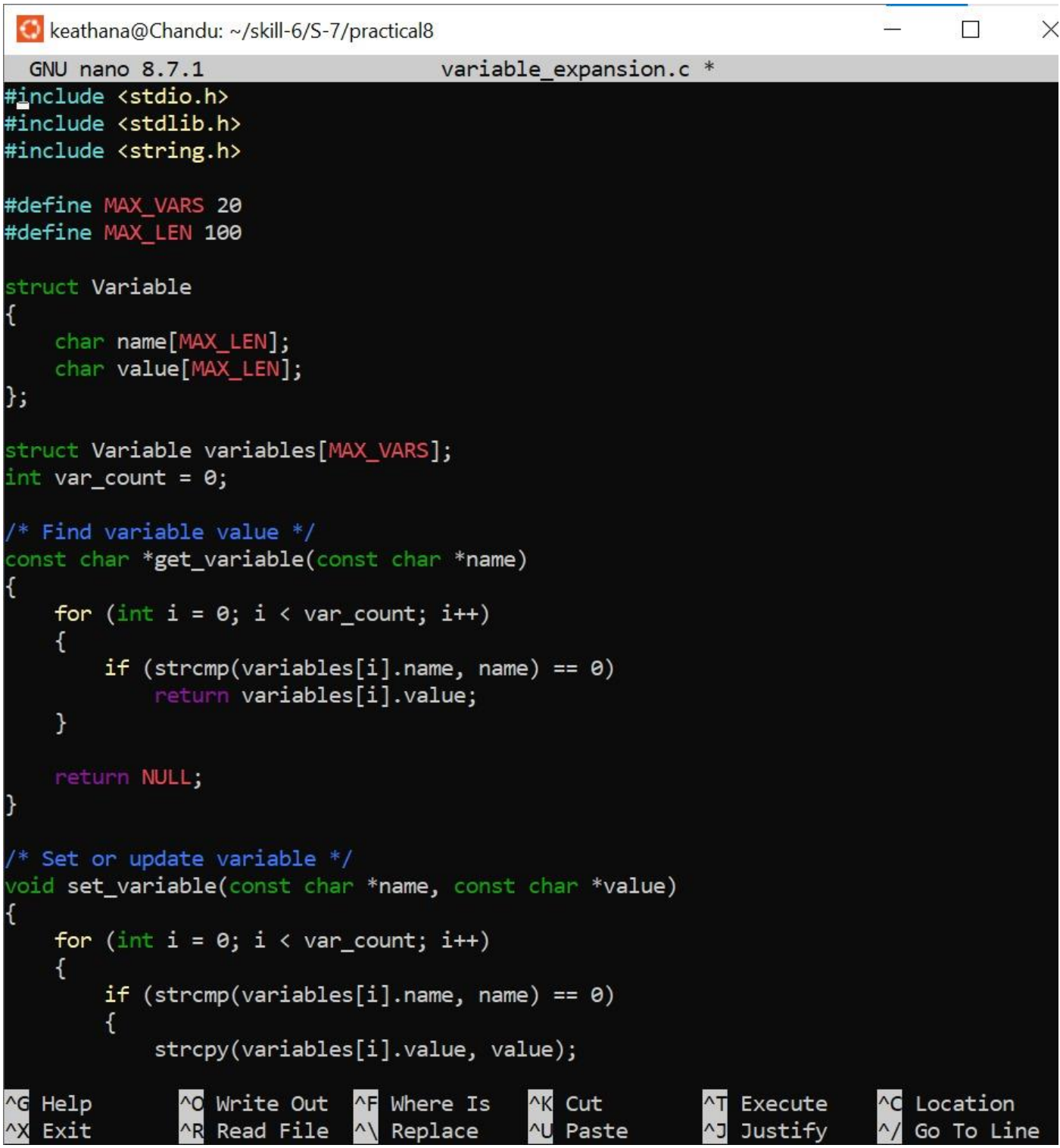
skill-8

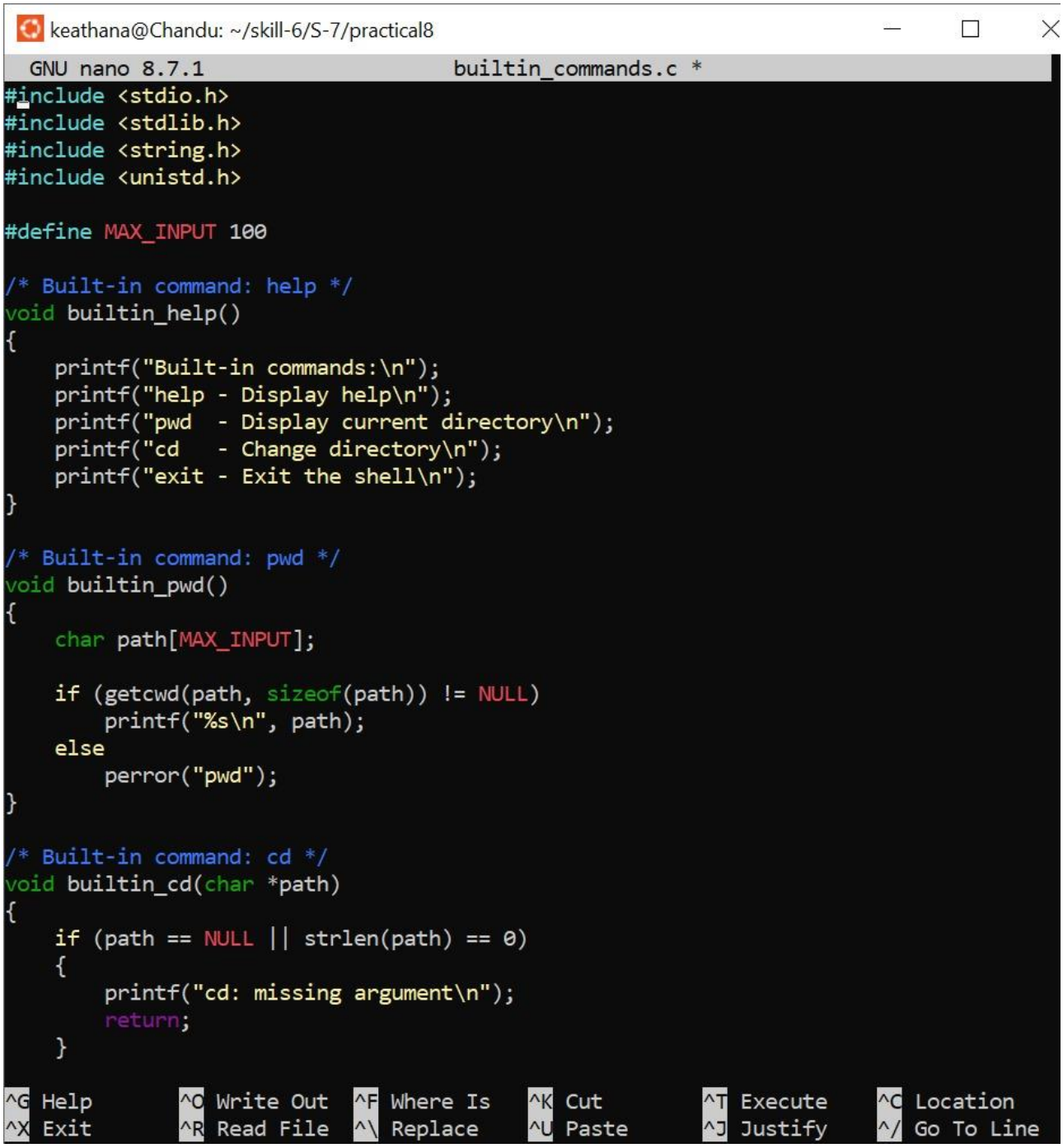
Input-

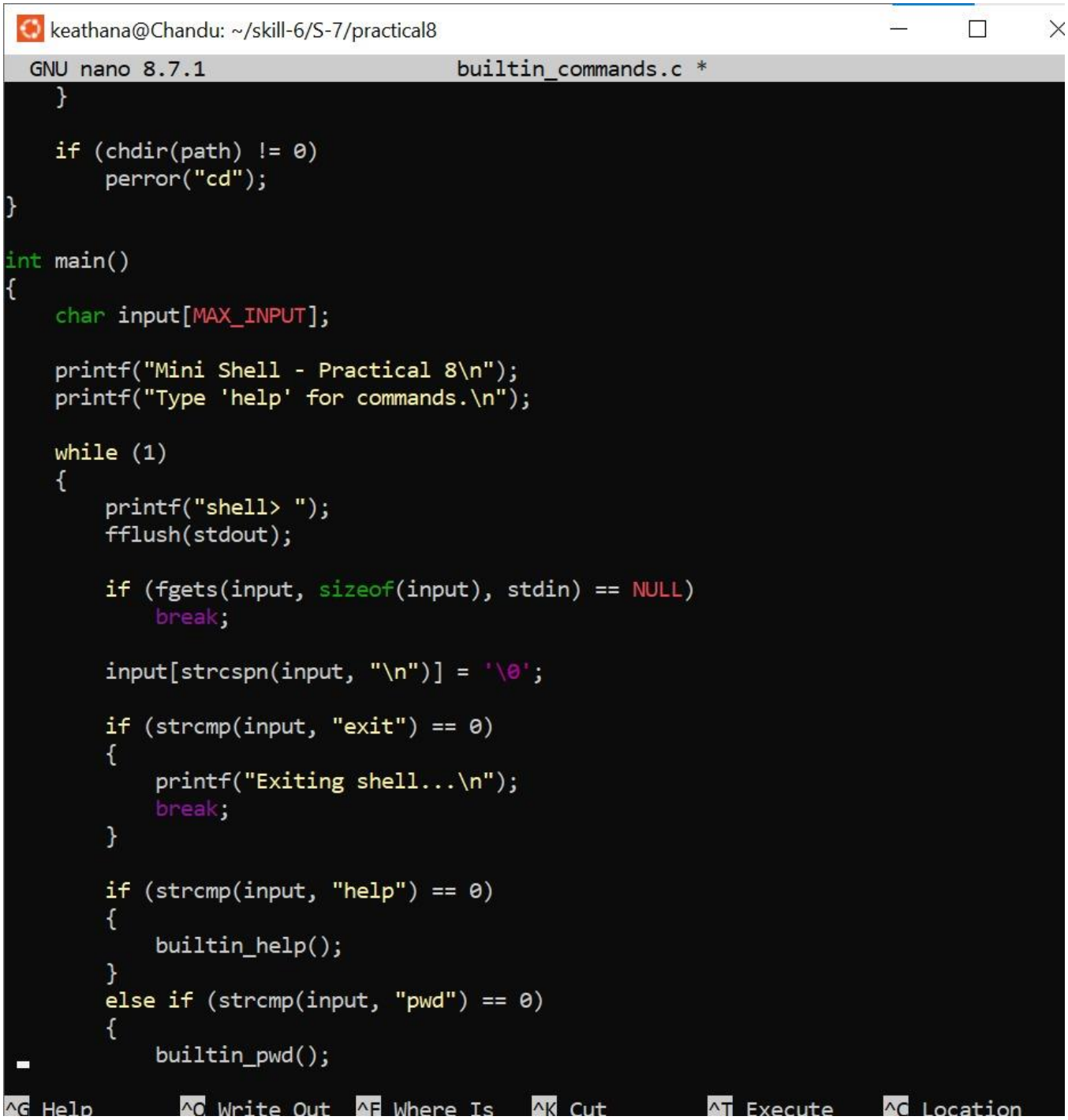



keathana@Chandu: ~/skill-6/S-7/practical8
GNU nano 8.7.1 variable_expansion.c *
return;
}
}
if (var_count < MAX_VARS)
{
strcpy(variables[var_count].name, name);
strcpy(variables[var_count].value, value);
var_count++;
}
}
/* Expand variables */
void expand_variables(const char *input, char *output)
{
int i = 0;
int j = 0;
while (input[i] != '\0')
{
if (input[i] == '$')
{
i++;
char name[MAX_LEN];
int k = 0;
while (input[i] != '\0' &&
((input[i] >= 'A' && input[i] <= 'Z') ||
(input[i] >= 'a' && input[i] <= 'z') ||
(input[i] >= '0' && input[i] <= '9') ||
input[i] == '_'))
{
name[k++] = input[i++];
}
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line


keathana@Chandu: ~/skill-6/S-7/practical8
GNU nano 8.7.1 variable_expansion.c *
}
name[k] = '\0';
const char *value = get_variable(name);
if (value != NULL)
{
strcpy(&output[j], value);
j += strlen(value);
}
else
{
/* Undefined variable becomes empty */
}
}
else
{
output[j++] = input[i++];
}
}
output[j] = '\0';
}
int main()
{
char input[MAX_LEN];
char output[MAX_LEN];
/* Create variables */
set_variable("NAME", "Hello Ubuntu");
set_variable("USER", "student");
set_variable("HOME", "/home/student");
printf("Variable Expansion Test\n");
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line

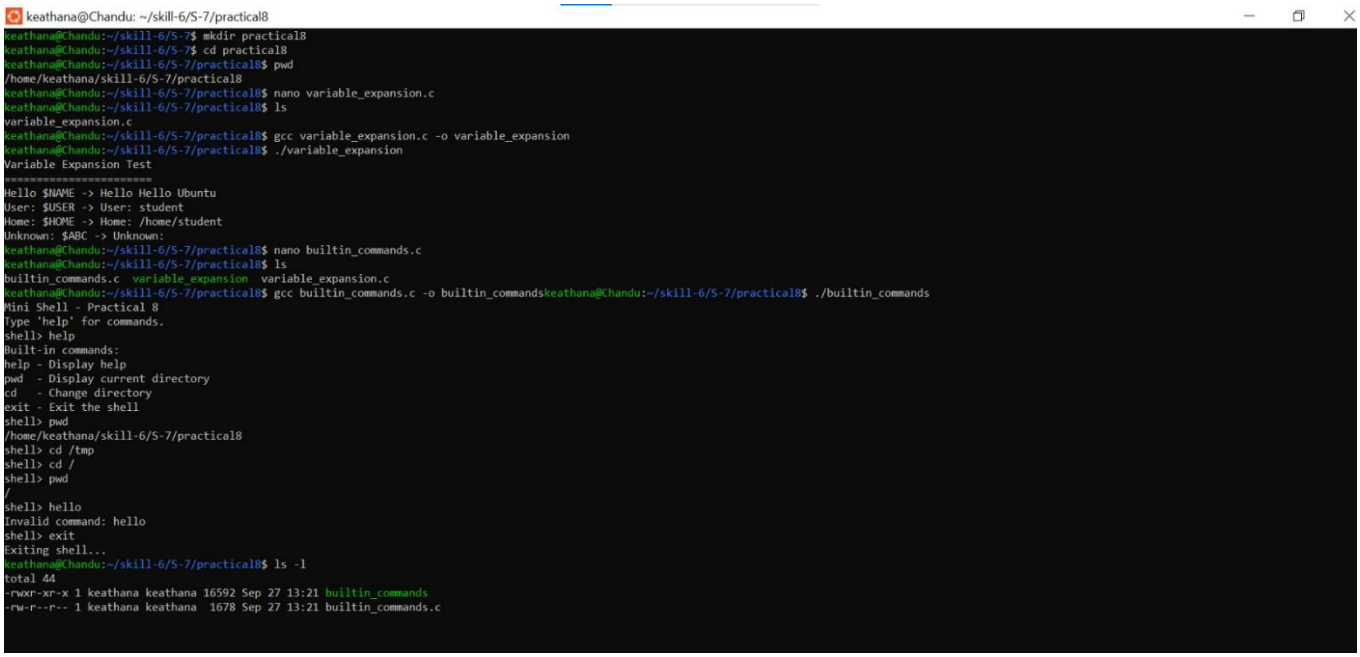

keathana@Chandu: ~/skill-6/S-7/practical8
GNU nano 8.7.1 variable_expansion.c *
output[j] = '\0';
}
int main()
{
char input[MAX_LEN];
char output[MAX_LEN];
/* Create variables */
set_variable("NAME", "Hello Ubuntu");
set_variable("USER", "student");
set_variable("HOME", "/home/student");
printf("Variable Expansion Test\n");
printf("======================\n");
strcpy(input, "Hello $NAME");
expand_variables(input, output);
printf("%s -> %s\n", input, output);
strcpy(input, "User: $USER");
expand_variables(input, output);
printf("%s -> %s\n", input, output);
strcpy(input, "Home: $HOME");
expand_variables(input, output);
printf("%s -> %s\n", input, output);
strcpy(input, "Unknown: $ABC");
expand_variables(input, output);
printf("%s -> %s\n", input, output);
return 0;
}
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line


keathana@Chandu: ~/skill-6/S-7/practical8
GNU nano 8.7.1 builtin_commands.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#define MAX_INPUT 100
/* Built-in command: help */
void builtin_help()
{
printf("Built-in commands:\n");
printf("help - Display help\n");
printf("pwd - Display current directory\n");
printf("cd - Change directory\n");
printf("exit - Exit the shell\n");
}
/* Built-in command: pwd */
void builtin_pwd()
{
char path[MAX_INPUT];
if (getcwd(path, sizeof(path)) != NULL)
printf("%s\n", path);
else
perror("pwd");
}
/* Built-in command: cd */
void builtin_cd(char *path)
{
if (path == NULL || strlen(path) == 0)
{
printf("cd: missing argument\n");
return;
}
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line


keathana@Chandu: ~/skill-6/S-7/practical8
GNU nano 8.7.1 builtin_commands.c *
}
if (chdir(path) != 0)
perror("cd");
}
int main()
{
char input[MAX_INPUT];
printf("Mini Shell - Practical 8\n");
printf("Type 'help' for commands.\n");
while (1)
{
printf("shell> ");
fflush(stdout);
if (fgets(input, sizeof(input), stdin) == NULL)
break;
input[strcspn(input, "\n")] = '\0';
if (strcmp(input, "exit") == 0)
{
printf("Exiting shell...\n");
break;
}
if (strcmp(input, "help") == 0)
{
builtin_help();
}
else if (strcmp(input, "pwd") == 0)
{
builtin_pwd();
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location


keathana@Chandu: ~/skill-6/S-7/practical8
GNU nano 8.7.1 builtin_commands.c *
if (fgets(input, sizeof(input), stdin) == NULL)
break;
input[strcspn(input, "\n")] = '\0';
if (strcmp(input, "exit") == 0)
{
printf("Exiting shell...\n");
break;
}
if (strcmp(input, "help") == 0)
{
builtin_help();
}
else if (strcmp(input, "pwd") == 0)
{
builtin_pwd();
}
else if (strncmp(input, "cd ", 3) == 0)
{
builtin_cd(input + 3);
}
else if (strlen(input) == 0)
{
continue;
}
else
{
printf("Invalid command: %s\n", input);
}
}
return 0;
}
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line

Output-


keathana@Chandu: ~/skill-6/S-7/practical8
keathana@Chandu:~/skill-6/S-7$ mkdir practical8
keathana@Chandu:~/skill-6/S-7$ cd practical8
keathana@Chandu:~/skill-6/S-7/practical8$ pwd
/home/keathana/skill-6/S-7/practical8
keathana@Chandu:~/skill-6/S-7/practical8$ nano variable_expansion.c
keathana@Chandu:~/skill-6/S-7/practical8$ ls
variable_expansion.c
keathana@Chandu:~/skill-6/S-7/practical8$ gcc variable_expansion.c -o variable_expansion
keathana@Chandu:~/skill-6/S-7/practical8$ ./variable_expansion
Variable Expansion Test
==========
Hello $NAME -> Hello Hello Ubuntu
User: $USER -> User: student
Home: $HOME -> Home: /home/student
Unknown: $ABC -> Unknown:
keathana@Chandu:~/skill-6/S-7/practical8$ nano builtin_commands.c
keathana@Chandu:~/skill-6/S-7/practical8$ ls
builtin_commands.c variable_expansion variable_expansion.c
keathana@Chandu:~/skill-6/S-7/practical8$ gcc builtin_commands.c -o builtin_commandskeathana@Chandu:~/skill-6/S-7/practical8$ ./builtin_commands
Mini Shell - Practical 8
Type 'help' for commands.
shell> help
Built-in commands:
help - Display help
pwd - Display current directory
cd   - Change directory
exit - Exit the shell
shell> pwd
/home/keathana/skill-6/S-7/practical8
shell> cd /tmp
shell> cd /
shell> pwd
/
shell> hello
Invalid command: hello
shell> exit
Exiting shell...
keathana@Chandu:~/skill-6/S-7/practical8$ ls -l
total 44
-rwxr-xr-x 1 keathana keathana 16592 Sep 27 13:21 builtin_commands
-rw-r--r-- 1 keathana keathana 1678 Sep 27 13:21 builtin_commands.c